TD03_Comparison_to_theoretical_mean

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1 TD 3: Comparison of sample mean to theoretical mean

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Goal of the session: Compare sample mean to theoretical mean

1.1 1. Data presentation

A clinician evaluates the intensity of depression on the Hamilton scale, which ranges from 0 (not depressed) to 56 (very depressed), in 100 patients suffering from a Major Depressive Episode (MDE). We want to know whether acupuncture improves the course of depression. To this end, 100 patients with the same depressive state (same Hamilton score) were divided into 2 groups of 50 by drawing lots. by lot. Group I received pharmacological treatment only, while Group II received pharmacological treatment combined with acupuncture. pharmacological treatment combined with acupuncture. The dataset contains the following variables: - `hamilton_g1`: Hamilton scores, after 8 months of treatment, for patients of group I; - `hamilton_g2`: Hamilton scores, after 8 months of treatment, for patients of group II; - `urbanization_g1`: level of urbanization in which patients live (A: rural, B: moderate urbanization, C: highly urbanized) for patients of group I; - `urbanization_g2`: level of urbanization in which patients live (A: rural, B: moderate urbanization, C: highly urbanized) for patients of group II; - `age_g1`: patients' age in years for group I; - `age_g2`: patients' age in years for group II; - `weight_g1`: patients' weight in kilograms for group I; - `weight_g2`: patients' weight in kilograms for group II; - `sex_g1`: patients' gender in group I; - `sex_g2`: patients' gender in group II.

The data is contained in the text file `data_depression.txt` in the `data/` directory, whose full path is `data/data_depression.txt`.

Data are read using R's `read.table()` function:

[1]:
```
# Read data
depression = read.table("data/data_depression.txt", header = TRUE)


# Display the beginning of the data file contents
head(depression)



# load the function you'll need (from now on, remember to copy-paste it␣
 ↪yourself in subsequent sessions)
```

```
C.I <- function(x,alpha){
    alpha <- 1-alpha
    mu = mean(x)
    sigma <- sd(x)
    n = length(x)
    CI_inf = mu - (qt(1 - alpha/2,n-1) * sigma/sqrt(n))
    CI_sup = mu + (qt(1 - alpha/2,n-1) * sigma/sqrt(n))
    CI = c(inf = CI_inf ,sup = CI_sup)
    return(CI)

}
```

A data.frame: 6 × 10

	hamilton_g1 <int>	hamilton_g2 <int>	urbanization_g1 <chr>	urbanization_g2 <chr>	age_g1 <int>	age_g2 <int>
1	43	25	B	B	57	48
2	29	29	A	B	37	41
3	34	18	A	A	7	30
4	39	23	A	A	37	41
5	40	24	B	A	15	30
6	41	25	B	B	45	45

1.2 2. Bilateral test

Objective:

We would like to know whether the Hamilton scores observed in Group 1 women women are equal on average to 29..

We need to perform a Student's t test comparing the mean estimated by the function function with the expected theoretical mean E = 29 in our case.

The principle of this test is simple: it checks whether the difference between the observed mean μ in the sample theoretical mean μ_0 is sufficiently far from 0 with a certain risk of error risk of error α for the two averages to be considered different. The test therefore calculates the difference between μ and μ_0 and divides by its standard deviation to normalize:

$$T = \frac{\mu - \mu_0}{\sqrt{\frac{S^2}{n}}} = \frac{mu - \mu_0}{\frac{S}{\sqrt{n}}}$$

H0: Hamilton score for women in group 1 is equal to 29.

H1: Hamilton score for women in group 1 is not equal to 29.

Sample extraction: - extract the actual sample - get back its size using the `length()` function

[3]:
```
# Data extraction
w1 = depression[depression$sex_g1 == "F", "hamilton_g1"]

# Display extracted data
w1
```

```r
# Sample size
length(w1)
```

1. 43 2. 37 3. 38 4. 40 5. 37 6. 41 7. 43 8. 47 9. 46 10. 42 11. 44

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Checking application conditions

Check sample size: - If the sample size is large ($n \geq 30$): no application condition - If sample size is small ($n < 30$): data must follow a normal distribution

Check application conditions

Hypotheses of the Shapiro test

H0 : the Hamilton scores of women in group I follow a normal distribution

H1 : the Hamilton scores of women in group I don't follow a normal distribution

[4]:
```r
# Shapiro test
shapiro.test(w1)
```

```
        Shapiro-Wilk normality test

data:  w1
W = 0.94764, p-value = 0.614
```

Conclusion :

We observe p = 0.61 »> alpha = 0.05, thus we cannot reject H0, so we conclude that the scores of women in group I follow a normal distribution

Student's t-test You can perform the test using the `t.test()` function which takes the sample as a first argument, and the theoretical mean `m` of value `X` as another argument `m = X` (Note: `X` here should be an actual number when calling the function).

[5]:
```r
# Student's t-test
t.test(w1, m = 29)
```

```
        One Sample t-test

data:  w1
t = 12.276, df = 10, p-value = 2.358e-07
alternative hypothesis: true mean is not equal to 29
95 percent confidence interval:
 39.34289 43.92984
sample estimates:
mean of x
```

```
41.63636
```

Conclusion :

We observe p = 2.4 10 «< alpha = 0.05, thus we reject H0, which means that the average Hamilton score of women in group I is different from 29.

1.3 3. One-sided test

Objective:

We would like to know whether the Hamilton scores observed in Group 1 women women are higher on average than 29.

In this case, we will perform a one-tailed test:

H0: the Hamilton score for women in group 1 is equal to 29.

H1: the Hamilton score for women in group 1 is greater than 29.

[6]:
```
# One-sided t-test
t.test(w1, m = 29, alternative = "greater")
```

```
        One Sample t-test

data:  w1
t = 12.276, df = 10, p-value = 1.179e-07
alternative hypothesis: true mean is greater than 29
95 percent confidence interval:
 39.77076      Inf
sample estimates:
mean of x
 41.63636
```

Conclusion:

We observe p = 1.2 10 «< alpha = 0.05, thus we reject H0, which means that the average Hamilton score of women in group I is greater than 29.

1.4 4. Exercises

1.4.1 Exercise 1

Answer the following questions, tests will be performed at threshold $\alpha = 0.02$:

1. Is the average Hamilton score for women in group 2 below 29?
2. Are the Hamilton scores observed in group 1 for men higher on average than 29?
3. Are the Hamilton scores observed in group 2 for men higher on average than 29?

Conclude.

[14]:
```r
## 1.
# Extract data
w2 = depression[depression$sex_g2 == "F", "hamilton_g2"]

# Sample size
length(w2)

# Check conditions
# Shapiro test with
#     H0: w2 scores follow a normal distribution
#     H1: w2 scores don't follow a normal distribution
shapiro.test(w2)

# => p = 0.83 >>> alpha = 0.02, so we can't reject H0, which means that w2␣
 ↪scores follow a normal distribution

# Perform one-sided t-test
# H0: w2 is equal to 29
# H1: w2 is less than 29
t.test(w2, m = 29, alternative = "less")

# We observe p = 0.08 > alpha = 0.02, so we can't reject H0, which means that␣
 ↪w2 is equal to 29

## 2.
m1 = depression[depression$sex_g1 == "H", "hamilton_g1"]

# Sample size
length(m1)

# n > 30, can proceed to the test directly
# H0: m1 = 29
# H1: m1 > 29
t.test(m1, m = 29, alternative = "greater")

# We observe p = 5.8 10 ⁻¹  <<< alpha = 0.02, so we reject H0, and m1 > 29

## 3.
m2 = depression[depression$sex_g2 == "H", "hamilton_g2"]

# Sample size
length(m2)
# n > 30, we can perform the test
# H0: m2 = 29
# H1: m2 > 29
t.test(m2, m = 29, alternative = "greater")
```

```
# We observe p = 1 >>> alpha = 0.02, so we can't reject H0, and m2 = 29
```

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```
        Shapiro-Wilk normality test

data:  w2
W = 0.96258, p-value = 0.8258
```

```
        One Sample t-test

data:  w2
t = -1.7233, df = 4, p-value = 0.07997
alternative hypothesis: true mean is less than 29
95 percent confidence interval:
     -Inf 29.99576
sample estimates:
mean of x
     24.8
```

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```
        One Sample t-test

data:  m1
t = 11.251, df = 38, p-value = 5.846e-14
alternative hypothesis: true mean is greater than 29
95 percent confidence interval:
 36.62952      Inf
sample estimates:
mean of x
 37.97436
```

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```
        One Sample t-test

data:  m2
t = -5.9973, df = 44, p-value = 1
alternative hypothesis: true mean is greater than 29
95 percent confidence interval:
 23.25349      Inf
sample estimates:
mean of x
```

```
24.51111
```

1.4.2 Exercise 2 (homework)

Answer the following questions, the tests will be performed at the threshold $\alpha = 0.01$:

1. Are the Hamilton scores observed in group 2 in rural areas different on average at 29?
2. Are the Hamilton scores observed in group 2 in moderately urbanized areas different on average at 29?
3. Are the Hamilton scores observed in group 2 in highly urbanized areas lower on average than 25?
4. Are the Hamilton scores observed in group 1 in rural areas below 39 on average?
5. Are the Hamilton scores observed in group 1 in moderately urbanized areas higher than 29 on average?
6. Are the Hamilton scores observed in group 1 in highly urbanized areas different from 25 on average?

Conclude.

[ ]:
```
# I'll let the student perform the tests
```